

FXR90 Antenna and Cable Degradation Predictor

A practical AI system that predicts antenna disconnects, cable degradation, RF path loss, and antenna misalignment using FXR90 reader telemetry and tag-read quality signals.

Antenna Health

Cable Loss

RSSI Trends

Read Quality

Per Port

Health scoring for each antenna/read point

Early

Detect gradual RF quality degradation

Actionable

Recommend cable, antenna, and placement checks

PROJECT SUMMARY

Predict RF path degradation before tag reads fail

Antennas and RF cables are exposed to vibration, movement, dust, physical damage, connector looseness, moisture, and installation drift. A reader may still be online while one antenna path slowly becomes weak. This project detects that early by comparing per-antenna read volume, RSSI, phase, channel behavior, connection state, and cable compensation settings over time.

Read Drop

Detect falling tag reads per antenna compared with baseline.

RF Quality

Track RSSI, phase consistency, channel spread, and read count.

Cable Risk

Use cable length and loss compensation as context.

Disconnect

Identify hard disconnects and intermittent antenna state changes.

Business Problem

- One weak antenna can reduce RFID coverage while the reader still appears healthy.
- Loose connectors and cable wear often show gradual RF decline before total failure.
- Manual inspection is slow when many dock doors, portals, or read zones exist.
- Operations need per-antenna alerts with clear field-service action.

AI Outcome

- Health score for each antenna/read point.
- Predicted issue type: disconnect, cable loss, misalignment, obstruction, or configuration mismatch.
- Confidence and trend explanation.
- Recommended inspection or corrective action.

Showcase Statement

The project turns normal FXR90 tag-read telemetry into an AI maintenance assistant that finds weak antennas and degrading RF cables before inventory accuracy or portal performance drops.

API SIGNALS

The API exposes direct and indirect antenna-health indicators

The model combines explicit connection status with RF performance signals. Direct state tells us whether an antenna is connected. Trend signals tell us whether a connected antenna is quietly becoming worse.

Signal Group	Useful Fields	Maintenance Meaning
Antenna State	<code>radio_control.antennas</code> <code>get_status.antennas</code>	Detect connected, disconnected, unknown, and intermittent state changes per antenna port.
Read Volume	<code>radio_control.numTagReads</code> <code>numTagReadsPerAntenna</code> <code>mode_tag_data_events.reads</code>	Find per-antenna read drops compared with historical baseline, neighboring antennas, or expected traffic.
RF Strength	<code>mode_tag_data_events.peakRssi</code> <code>antenna</code> <code>idHex</code>	Detect signal attenuation, poor antenna aim, damaged cable, connector loss, or blocked read zone.
RF Behavior	<code>phase</code> <code>channel</code> <code>eventNum</code>	Measure stability and consistency of tag reads. Sudden variance changes can indicate environmental or antenna-path problems.
RF Configuration	<code>get_mode.antennas</code> <code>transmitPower</code> <code>environment</code> <code>tagMetaData</code>	Separate true degradation from changed mode, reduced power, disabled antenna, or changed RF environment profile.
Cable Context	<code>get_cableLossCompensation</code> <code>cableLength</code> <code>cableLossPerHundredFt</code>	Use expected cable attenuation as a baseline and detect mismatches between configured compensation and observed RF quality.
Diagnostics	<code>warning.message</code> <code>error.message</code> <code>numWarnings</code> <code>numErrors</code>	Use reader diagnostics to support root cause, especially antenna disconnects and radio faults.

Important Measurement Note

RSSI and read counts are affected by tag population, product mix, antenna placement, reader mode, transmit power, and environment. The model should compare each antenna against its own historical baseline and current operating configuration, not a single global threshold.

PREDICTION TARGETS

Classify the most likely antenna or cable problem

The system should not only say “bad antenna.” It should estimate the most likely issue type so the technician knows where to start.

Predicted Failure Classes

- **Hard disconnect:** antenna status changes to disconnected or unknown.
- **Intermittent cable/connector:** connection flips or RF strength drops in bursts.
- **Gradual cable loss:** RSSI and read rate decline slowly over days or weeks.
- **Antenna misalignment:** read zone becomes weaker but connection remains stable.
- **Obstruction/environment change:** one zone degrades after layout or material change.
- **Configuration mismatch:** read drop caused by changed mode, disabled port, or transmit power.

Example Alert

Antenna 3 degradation risk is High.

Read volume dropped 42% versus the 14-day baseline, median RSSI is 6.8 dB weaker, and transmit power/configuration did not change. The most likely cause is cable loss, loose connector, or antenna misalignment.

Action: inspect Antenna 3 cable path, tighten connectors, check antenna mounting angle, then rerun validation scan.

Symptom Pattern	Likely Root Cause	Evidence Used
Antenna state reports disconnected.	Physical disconnect, failed cable, failed antenna, failed port.	<code>radio_control.antennas</code> , <code>get_status.antennas</code> , error/warning messages.
RSSI declines while read volume declines.	Cable degradation, connector looseness, antenna damage, antenna moved away.	<code>peakRssi</code> , <code>reads</code> , <code>numTagReadsPerAntenna</code> .
Read volume drops but RSSI remains normal.	Traffic/tag population change, obstruction, workflow change, zone usage change.	Reads by antenna, expected operating schedule, neighboring antenna comparison.
RSSI is weak after known cable setting change.	Incorrect cable loss compensation or cable metadata mismatch.	<code>get_cableLossCompensation</code> , <code>set_cableLossCompensation</code> audit, RSSI baseline.
All antennas degrade together.	Reader config, RF environment, reader power, tag population, or site-wide interference.	<code>get_mode</code> , transmit power, environment profile, radio state, warnings.

Best MVP Target

Start with per-antenna degradation detection for connected antennas: read-rate drop, RSSI drop, and no matching configuration change. This gives high operational value and avoids overcomplicating the first release.

AI LOGIC

Score each antenna using trend, quality, and configuration context

A useful model needs to avoid false alarms caused by normal traffic changes. The score should blend antenna connection status, read-volume trend, RF signal trend, variance, and configuration checks.

Score Components

Connection state		30%
Read-rate drop		25%
RSSI degradation		25%
RF variance		10%
Config mismatch		10%

```
AntennaDegradationScore =  
0.30 * connection_risk  
+ 0.25 * read_rate_drop_risk  
+ 0.25 * rssi_drop_risk  
+ 0.10 * rf_variance_risk  
+ 0.10 * config_mismatch_risk
```

Risk Level:

0-39 Low | 40-64 Medium
65-84 High | 85-100 Critical

Baseline Features

- Median RSSI per antenna per hour.
- Tag reads per antenna per operating window.
- Unique tag count per antenna.
- RSSI percentiles and trend slope.

Comparison Features

- Antenna vs its own 7/14/30 day baseline.
- Antenna vs neighboring antennas.
- Current RF config vs previous config.
- Expected zone activity vs actual activity.

Diagnostic Features

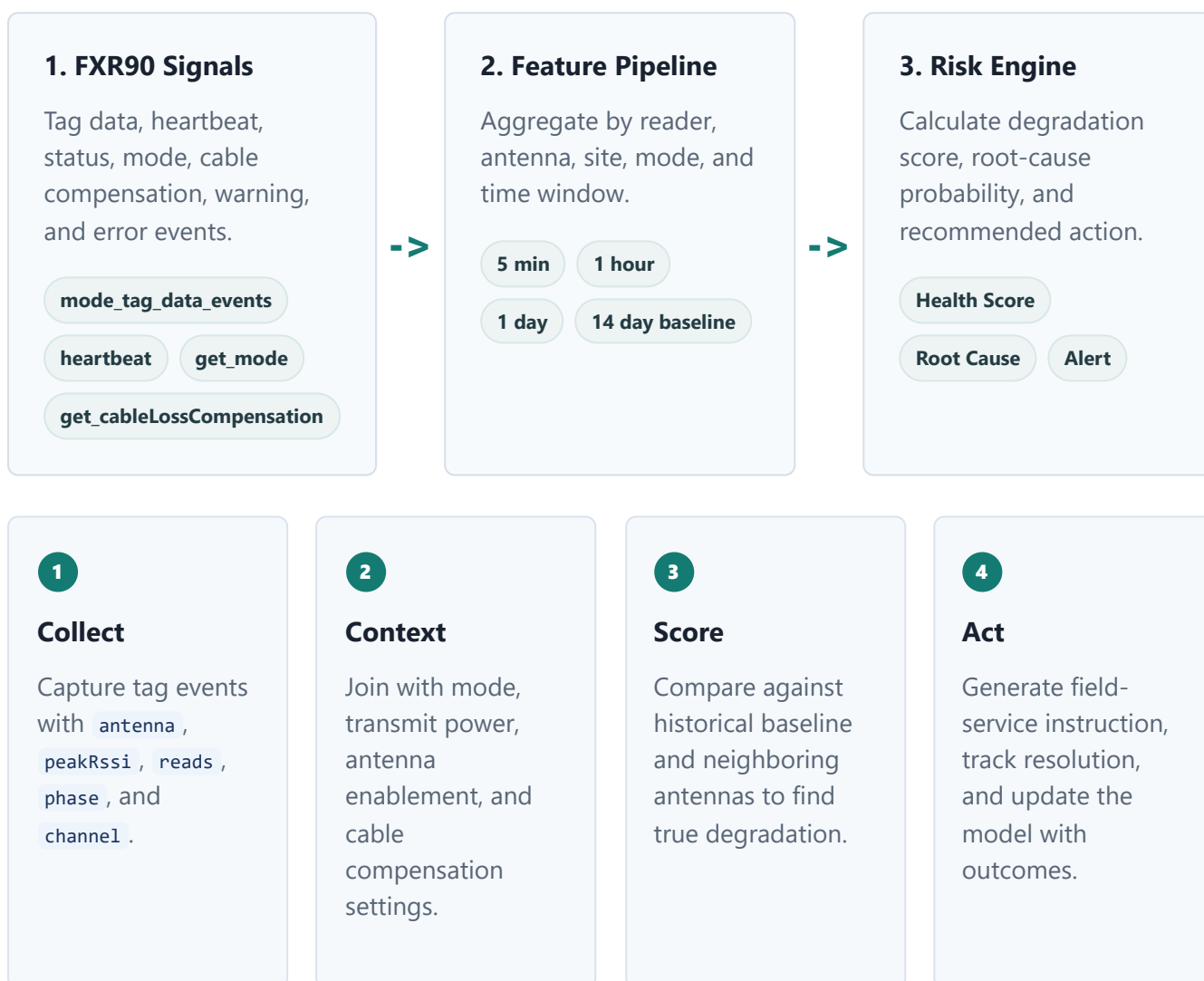
- Connection flips per day.
- Warning and error message counts.
- Radio connection health.
- Cable length and loss metadata.

Model Stage	Recommended Method	Reason
MVP	Rules plus per-antenna baseline anomaly detection.	Works without labeled failure history and is easy to explain to operations.
Phase 2	Isolation Forest, robust z-score, or EWMA trend detection.	Good for discovering unusual drops and slow degradation in time-series telemetry.
Phase 3	Supervised classifier using service records and confirmed root causes.	Improves root-cause classification after maintenance feedback is collected.

ARCHITECTURE

From tag-read events to antenna maintenance actions

The system listens to tag data and management events, normalizes them by reader and antenna, creates per-port feature windows, and publishes health scores with maintenance recommendations.



Data Tables

- **raw_tag_events:** reader, antenna, tag ID, RSSI, phase, channel, reads, timestamp.
- **antenna_features:** rolling metrics per antenna and time window.
- **reader_config_snapshots:** mode, active antennas, transmit power, environment.
- **cable_profiles:** cable length and loss compensation by read point.
- **maintenance_actions:** inspection results and root-cause labels.

Model Output

- **reader_id:** FXR90-01
- **antenna:** 3
- **health_score:** 41 / 100
- **risk_level:** High
- **likely_issue:** cable/connector degradation
- **evidence:** RSSI drop and read-rate drop with no config change

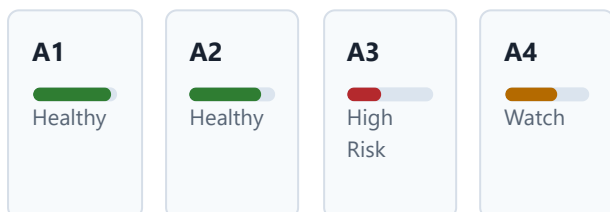
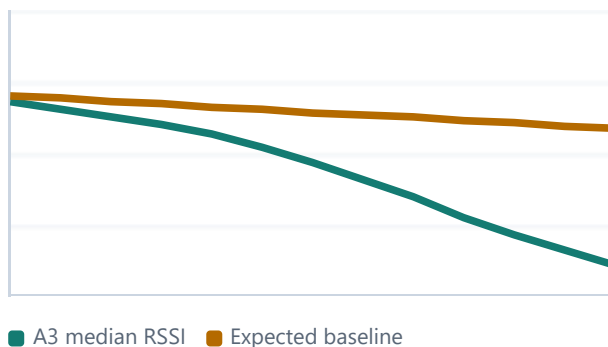
USER EXPERIENCE

Show per-antenna health in a way technicians can act on

The dashboard should make it easy to identify which antenna is weak, why the model thinks so, and what to inspect first.

Reader FXR90-01 | Portal A

Mode: INVENTORY | Active antennas: 1-4 | Last event: 18 seconds ago

Antenna 3 High Risk**Antenna Health****Antenna 3 RF Trend****Sample Alert****Antenna 3 cable degradation suspected.**

Median RSSI is 6.8 dB lower than the 14-day baseline, read volume is down 42%, and the antenna is still reported as connected. Mode, transmit power, and active antenna configuration are unchanged.

Action: inspect cable run and connectors, check antenna mounting angle, verify cable loss compensation, and perform validation scan.

Dashboard Widgets

- Reader map with per-antenna health color.
- RSSI trend by antenna.
- Read-count trend by antenna.
- Connection state history.
- Cable compensation values by port.
- Evidence list and recommended action.

Risk Level	Meaning	Recommended Action
Low	Antenna quality is within normal baseline.	Continue monitoring.
Medium	Early RF drop or read imbalance detected.	Review during next routine inspection.
High	Sustained RSSI/read drop or intermittent connection pattern.	Inspect antenna and cable within the same shift.
Critical	Antenna disconnected or near-total read failure.	Immediate inspection and service task.

DELIVERY PLAN

Start with explainable detection, then improve with maintenance feedback

The first release should focus on connected antennas that show abnormal RF quality decline. After technicians confirm root causes, those outcomes become labels for better ML classification.

1

Configure

Ensure tag metadata includes antenna, RSSI, reads, and phase/channel where possible.

2

Ingest

Collect tag events, heartbeat antenna status, mode, and cable loss configuration.

3

Baseline

Create normal profiles per reader, antenna, site, mode, and time window.

4

Alert

Generate risk, evidence, likely issue, and maintenance action.

5

Learn

Use inspection outcomes to tune thresholds and train root-cause models.

MVP Scope

- Per-antenna health score.
- Disconnected/unknown antenna alerts.
- Read-rate drop detection by antenna.
- RSSI degradation detection by antenna.
- Configuration-change guardrail using mode and transmit power.
- Cable loss compensation visibility by read point.
- Dashboard with recommended technician action.

Future Enhancements

- Root-cause classifier for cable, connector, antenna, obstruction, and configuration issues.
- Automated validation scan workflow after maintenance.
- Site map view with antenna coverage zones.
- Service ticket integration.
- Automatic cable compensation recommendation for engineer approval.
- Combined reader health plus antenna health scoring.

Phase	Deliverable	Success Metric
Week 1-2	Tag-event ingestion, antenna status ingestion, mode snapshot, cable profile table.	95%+ events mapped to reader and antenna.
Week 3-4	Baseline features and first antenna degradation score.	Detect known weak/disconnected antennas in pilot data.
Week 5-6	Dashboard, alert feed, inspection workflow, and technician feedback capture.	Each alert includes evidence and action.
After MVP	Anomaly tuning, root-cause classifier, and maintenance-system integration.	Reduced missed reads and faster antenna issue resolution.

BUSINESS VALUE

Why this project is worth building

Antenna and cable issues directly affect RFID read accuracy. This project catches weak read paths before they become operational misses, and it gives technicians precise evidence instead of asking them to inspect the whole site.

Higher Read Accuracy

Prevent blind spots and weak zones from growing unnoticed.

Faster Maintenance

Tell technicians exactly which antenna or cable path to inspect.

Lower Downtime

Fix degradation before read operations or portal workflows fail.

KPIs to Track

- Missed read rate by portal or zone.
- Number of antenna/cable incidents detected early.
- Mean time to identify weak antenna path.
- Mean time to repair antenna/cable issues.
- False positive and false negative alert rate.
- Post-maintenance RSSI/read-rate recovery.

Stakeholder Pitch

We can use FXR90 antenna status and tag-read telemetry to detect weak antennas and cable degradation before inventory accuracy suffers. The MVP is explainable, practical, and directly tied to operational reliability.

Recommended Next Step

Pilot on 3-5 readers with known antenna layouts. Capture at least two weeks of tag events and heartbeat status, build the per-antenna baseline, and validate alerts with field inspection results.

Decision Point	Recommendation
First prediction target	Connected antenna with sustained read-rate and RSSI decline.
First dashboard	Reader list, antenna grid, degradation score, RSSI trend, read trend, evidence, and action.
First model	Rules plus baseline anomaly detection by reader, antenna, mode, and time window.
Expansion	Add root-cause classification, service ticket integration, and validation scans.

Prepared as a project concept based on the FXR90 MQTT API documentation in this workspace.